

VISHNU BHUVANACHANDRAN

Mechanical CAD Technician | SolidWorks | Low-Voltage Power Systems | R&D Design
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PROFESSIONAL SUMMARY

Mechanical engineer with 10+ years of hands-on experience in 3D CAD design, low-voltage electro-mechanical systems, and prototype development. Proficient in SolidWorks with a strong track record of producing detailed 3D models, GD&T drawings, BOMs, and design documentation aligned with engineering standards. Experienced in operating rapid-prototyping facilities including 3D printing, CNC, and laser cutting, and in coordinating with cross-functional teams through the full product development lifecycle. Holds a PhD in Mechanical Engineering and an MSc in Data Analytics. Based in Dundalk and immediately available.

KEY MECHANICAL DESIGN & PROJECT WORK

EV Battery Thermal Management System

- Full design cycle in SolidWorks: 3D modelling, stack-up analysis, material selection, DFM/DFA review, prototype build, and verification testing.
- Developed assembly and detail drawings with tolerance analysis to minimise manufacturing risk and simplify assembly.

Industrial Central Vacuum System

- Collaborated with suppliers for component sourcing and BOM creation; produced manufacturable drawings and revision-controlled document packs.
- Managed timely responses to engineering information and drawing requests throughout the project lifecycle.

Electro-Mechanical Modular Prototyping

- Designed and developed modular prototypes with precise 3D CAD and GD&T drawings, analysing thermal and electro-mechanical requirements.
- Maintained Design History File (DHF)-style documentation and managed rapid iteration cycles across a multi-technology lab environment.

Welding Fume Reduction Safety Device

- Designed a fume-reduction helmet attachment (patent granted); performed CAD, prototyping, and validation against ISO 15011-1:2009 and AWS F1.2 standards.
- Produced risk assessments and full regulatory documentation package for submission.

Thermoelectric EV Battery Cooling Concept

- Designed a dual-stage thermoelectric cooling system with CAD modelling, thermal simulation, prototype build and test.
- Resulted in a granted patent and Best Paper Award at international conference.

PROFESSIONAL EXPERIENCE

Data Analyst

Jun 2025 – Present

Starilum Ltd | Drogheda, Ireland

- Manage and validate complex datasets using SQL, Excel, and Python, improving data quality for research-driven decision-making.
- Developed Power BI dashboards that improved decision accuracy by 40%.
- Integrated machine learning models to enhance commercial forecasting accuracy by 25%.

Research Lead & Assistant Nodal Officer (IEDC)

Sep 2019 – Aug 2024

Marian Engineering College | Kerala, India

- Delivered modular electro-mechanical projects using precise 3D CAD, GD&T drawings, and prototypes, maintaining DHF-style documentation throughout the design process.
- Managed a rapid prototyping lab (3D printing, CNC, laser cutting), accelerating design iteration cycles and retiring technical risks at early project stages.
- Designed SolidWorks assemblies and produced complete BOMs and design deliverable packs for stakeholder review.
- Coordinated with engineers and external suppliers on component sourcing, revision control, and drawing-pack management.

- Co-founded the Energy Research Lab and Makers Lab, boosting research and prototyping capacity by 80%.

Senior Research Fellow

May 2014 – Dec 2019

Indira Gandhi Centre for Atomic Research (IGCAR) | Tamil Nadu, India

- Designed and validated mechanical and electro-mechanical components to Government of India safety standards; executed Design of Experiments (DoE) programmes.
- Built and tested prototypes; performed failure analysis and implemented corrective actions through structured change control processes.
- Authored patents, risk assessments, peer-reviewed publications, and regulatory documentation.
- Co-led setup of a green consumables facility targeting low-emission welding processes aligned with occupational safety standards.

EDUCATION

MSc, Data Analytics – President's Gold Medal Winner

2024 – 2025

Dundalk Institute of Technology, Ireland

PhD, Mechanical Engineering (Occupational Health & Safety)

2014 – 2020

NIT Tiruchirappalli, India

M.Tech, Thermal Science

2011 – 2013

University of Kerala, India

B.Tech, Mechanical Engineering

2005 – 2009

University of Kerala, India

CORE TECHNICAL SKILLS

CAD & Design:

SolidWorks, 3D Modelling, GD&T Drawings, BOM Creation, DFM/DFA, Assembly Design, Panel Layouts

Prototyping & Manufacturing:

3D Printing, CNC Machining, Laser Cutting, Rapid Prototyping, Failure Analysis, Design of Experiments (DoE)

Standards & Documentation:

ISO 13485, PLM / ECR / ECO, DHF Documentation, Risk Assessment, Revision Control, ISO 15011-1:2009, AWS F1.2

Software & Data:

Power BI, Python, SQL, MS Office Suite, Minitab, Neo4j

CERTIFICATIONS & STANDARDS

- ISO 13485 – Quality Management System for Medical Devices (Udemy)
- PLM Fundamentals – Items, BOM, revisions, release workflows, ECR/ECO concepts (Udemy)
- ISO 15011-1:2009 / 10882-1 / AWS F1.2 – Welding Emission Characterisation
- Advanced MS Office – Word, Spreadsheets, PowerPoint (Skill Nation)
- Advanced Microsoft Power BI (Skill Nation)

AWARDS, PATENTS & PUBLICATIONS

- President's Gold Medal – First Class Honours, MSc Data Analytics, Dundalk IT (2025)
- Senior Research Fellowship – Indira Gandhi Centre for Atomic Research, Government of India
- Best Paper Award – Dual-Stage Thermoelectric Cooling Method for EV Batteries
- 4 Granted Patents + 1 Published – welding flux compositions, thermoelectric EV battery cooling, fume-reduction helmet, thermal-spray attachments
- 15+ Peer-Reviewed Publications – design, process safety, and manufacturing (full list available on request)